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Lab Report

ECE 2031 L04

03 September 2025

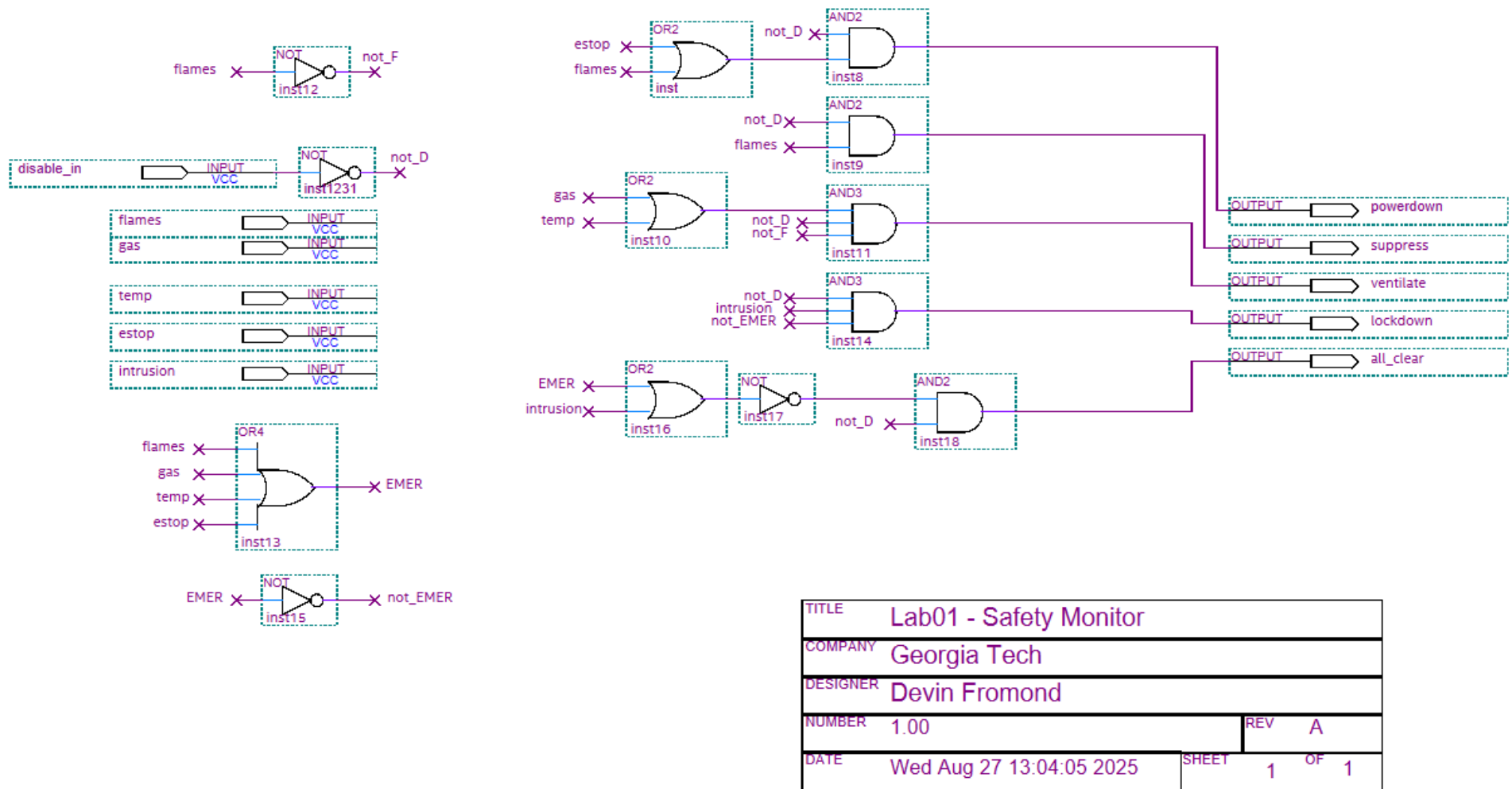


Figure 1. Schematic for the logical expressions pertaining to a safety-logic module with the outputs of powerdown ($/D \cdot (E + F)$), suppress ($/D \cdot F$), ventilate ($/D \cdot F \cdot (G + T)$), lockdown ($/D \cdot I \cdot EMER$), and all_clear ($/D \cdot /(EMER + I)$), where $EMER = F + G + T + E$ and the state abbreviations are the first letter of the state as pictured on the left of the diagram.

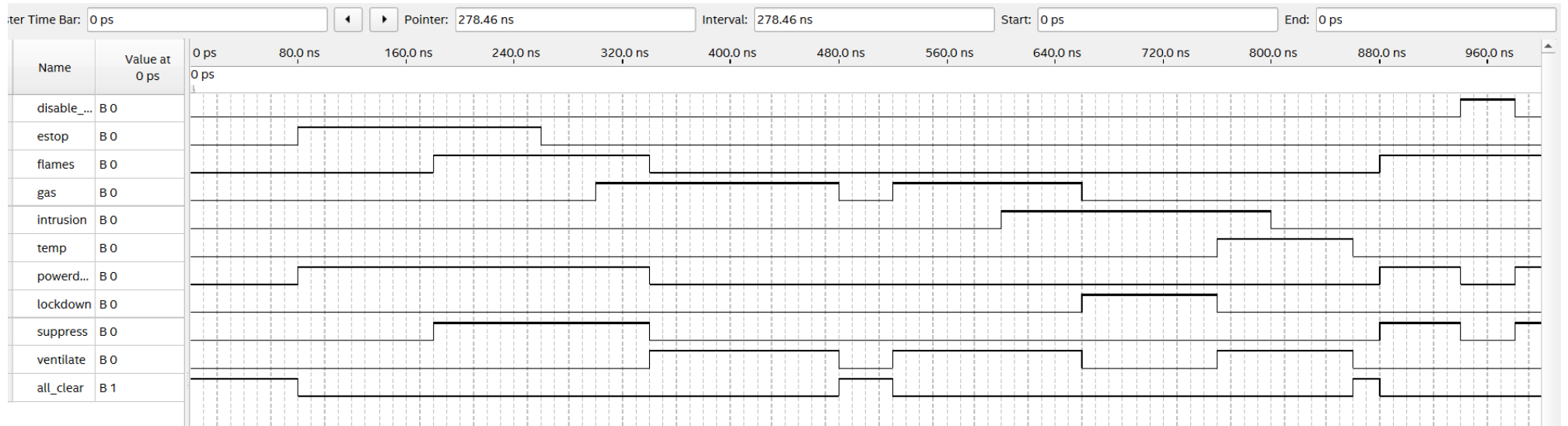


Figure 2. Waveform testing several conditions of the safety-logic module with their corresponding outputs.

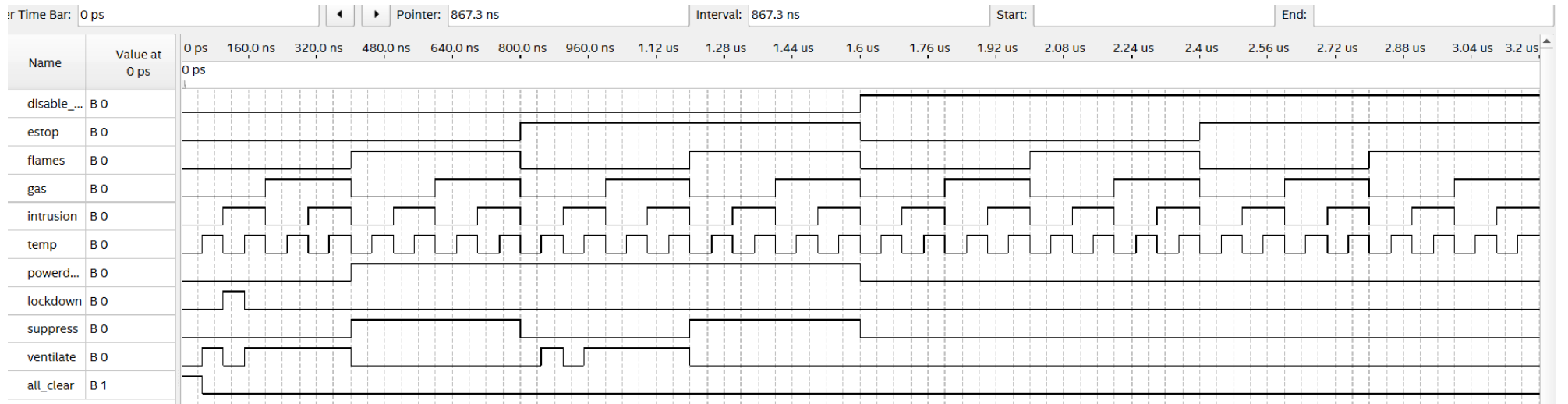


Figure 3. Waveform utilizing decreasing binary periods to test all conditions of the safety-logic module

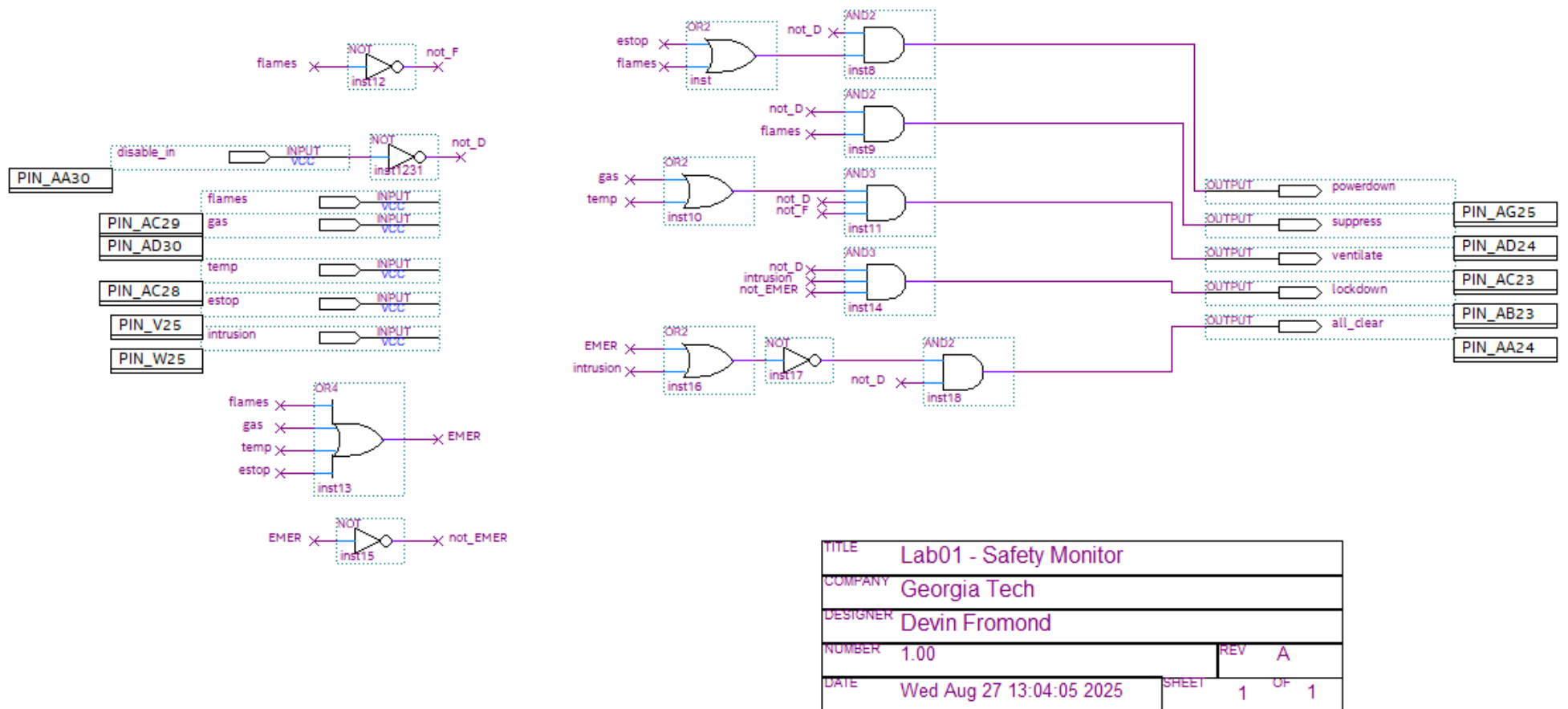


Figure 4. Schematic containing pins for an FPGA with designation 5CSXFC6D6F31C6 (identical logic to Figure 1.).